# Huawei Access Node QoS/Congestion Event Troubleshooting and Resolution SOP

**Node:** HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_2742

## Description

The QoS/Congestion Event failure domain, typified by microbursts, interface tail drops, and QoS queue starvation, on Huawei Access equipment situated at Hebbal Edge (Node ID: HEB-ACC-40), arises from intricate interactions between packet scheduling algorithms, traffic policing, and queuing mechanisms. Specifically, the Huawei equipment's QoS implementation, based on the DiffServ model, can lead to congestion when the aggregate traffic rate exceeds the interface's capacity, causing packets to be dropped. This phenomenon is exacerbated by the bursty nature of modern network traffic, where brief, high-volume transmissions can overwhelm the buffer capacity of network interfaces. The underlying protocol mechanics, including the Weighted Random Early Detection (WRED) and Priority Queuing (PQ), play a crucial role in managing congestion. However, misconfiguration or inadequate provisioning of these mechanisms can precipitate QoS queue starvation and interface congestion, triggering ALARM_2742. Understanding these complexities is essential for effective troubleshooting and resolution.

## Pre-requisite Checks

- Verify redundant link status to ensure no single point of failure
- Confirm availability of remote hands at Hebbal Edge for potential hardware interventions
- Check environmental conditions (temperature, power supply) at the site

## Potential Root Causes

| Cause | Probability |
|---|---|
| Inadequate bandwidth provisioning | High |
| Misconfigured QoS policies | Medium |
| Hardware failure (e.g., faulty interface card) | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display qos interface, Expected Output: QoS policy details and interface statistics, Description: This command isolates QoS configuration and performance issues on the affected interface, providing insights into packet scheduling, policing, and queuing behavior.

Command: display interface brief, Expected Output: Interface status, including errors and utilization, Description: Provides a snapshot of interface health, helping to identify potential issues such as high error rates or overutilization that could contribute to congestion.
```

```
Command: display traffic-statistics interface, Expected Output:
Detailed traffic statistics, including packet and byte counts,
Description: Offers in-depth traffic analysis, enabling the
identification of traffic patterns and potential sources of
congestion.
```

## Resolution Steps

Step 1: Step: Adjust QoS Policy, Command: qos policy class shaping , Description: Modify the existing QoS policy to adjust shaping parameters and ensure that traffic is properly policed, thereby preventing congestion due to excessive traffic rates.

Step 2: Step: Enable Priority Queuing, Command: qos policy class priority , Description: Configure priority queuing to ensure critical traffic is handled promptly, reducing the likelihood of QoS queue starvation and associated congestion issues.

Step 3: Step: Increase Interface Bandwidth, Command: interface bandwidth , Description: Upgrade the interface bandwidth to accommodate increased traffic demands, thus mitigating congestion and ensuring smooth network operation.

## Post-Check Verification

```
Command: display qos interface, Expected Output: Updated QoS policy
details reflecting adjustments, Description: Verifies that QoS policy
changes have been applied correctly and are functioning as expected.

Command: display interface brief, Expected Output: Improved interface
utilization and reduced errors, Description: Confirms that interface
performance has improved following the adjustments, with lower error
rates and healthier utilization levels.
```

## Roll-back Procedures

- Step: Revert QoS Policy, Command: undo qos policy , Description: Reverts QoS policy changes to their previous state in case the new configuration introduces unforeseen issues or fails to resolve the congestion problem.

- Step: Disable Priority Queuing, Command: undo qos policy class priority, Description: Removes priority queuing configuration if it was found to be ineffective or caused additional problems, returning to the original queuing behavior.

- Step: Revert Interface Bandwidth, Command: undo interface bandwidth, Description: Reverts the interface bandwidth to its original value if the increase did not alleviate congestion or introduced other performance issues.